# Statement of Research Interest

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I am a final-year MBBS student at AIIMS New Delhi working at the intersection of multimodal machine learning and healthcare deployment. Alongside my research, I cofounded Aterenex with a college junior, a health tech venture solving longitudinal patient electronic health record (EHR) fragmentation in Indian healthcare systems and improving insurance claim processing, work I will return to in more detail below. My core research conviction is that no single data modality, whether genomic, transcriptomic, proteomic, exposomic, histopathological, or radiological, is sufficient to characterize disease state, and that the highest value modeling problems in medicine are fundamentally multimodal representation learning problems: how to fuse heterogeneous, differently structured, differently noised signals into a shared latent space that is both predictive and clinically interpretable.

My technical work spans all these diverse modalities. In transcriptomics, I spent last summer at Harvard Medical School's MIND Data Science Lab under Dr. Sudeshna Das, working on single cell RNA transcriptomics for a project titled “Deciphering Astrocyte Transcriptomic Signature in Alzheimer's Disease and Tauopathies,” using Seurat, CellChat, and NicheNet to characterize astrocyte neuron interaction networks at single cell resolution. This work earned me a competitive travel fellowship to present as an oral presentation at the Alzheimer's Association Conference 2026 in London this July.

Building on this, I moved into computational pathology, working with a research team at Emory University under Dr. Anant Madabhushi on a project analyzing whole slide histopathology images in diffuse large B cell lymphoma. I used CellViT for deep learning based cell segmentation, then engineered a transition zone band framework to model spatial lymphocyte infiltration as a parabolic function around the tumor border, treating spatial topology, not just density, as a learnable feature for survival prediction. This work was selected for an oral presentation at the United States and Canadian Academy of Pathology (USCAP) 2026 conference in San Antonio, Texas.

I then extended into pure imaging based classification and prediction, continuing with the Madabhushi Lab on a project examining whether brain and retinal biological aging clocks are concordant. This involved training two independent deep learning models, a structural MRI derived brain age model and a foundation model based retinal age model using fundus imaging, across multiple large scale datasets including ADNI, Emory, BRSET, and UK Biobank, then applying residualized regression, interaction modeling, and spline based nonlinear analysis to test whether cross modal aging signals converge or diverge across chronological age. We are now extending this work to incorporate a proteomic aging data model as well, aiming to capture cross modality aging signal across imaging and molecular domains simultaneously. Alongside this, I trained a MobileNet based transfer learning classifier for automated Alzheimer's disease staging from structural brain MRI, using Grad-CAM for interpretability validation, and built a 1D ResNet on raw 12 lead ECG waveforms to predict cardiovascular biological age gap.

Returning to Aterenex, we are a group of five undergraduates from AIIMS Delhi and IIT Delhi working on two problems. First, longitudinal patient data integration: because EHR systems in India are fragmented and penetrance of the Ayushman Bharat Digital Mission is low, most patient data exists as disconnected episodic records, so we are building EMR systems that generate structured longitudinal patient timelines, a record linkage problem that is a prerequisite for any downstream ML in this setting. Second, insurance claim inefficiency: we are training a model to estimate expected claim amounts and predict the probability of claim acceptance, with recommendations to reduce rejection risk, a calibrated structured prediction problem.

This combination of training multimodal biomedical models across transcriptomic, histopathological, imaging, and now proteomic data, alongside building infrastructure to deploy such models in a resource constrained health system, motivates my interest in your Multimodal track. I am particularly interested in how large scale multimodal foundation model architectures and cross modal self supervised pretraining could be adapted to biomedical settings where labeled data is scarce and modalities are asynchronous, a problem I encounter directly through Aterenex's fragmented EHR data. I see this symposium as an opportunity to learn state of the art multimodal architectures from researchers operating at a scale I have not yet had access to.